Learning Outcomes

At the end of this exercise, students are capable to:

1. translate a data model (ER Diagram) to a relational database in mySQL.
2. capture existing data from a spreadsheet file to script in mySQL.
3. create tables and insert data through scripts to a relational database using phpmyadmin.

Designing the VSU Departmental Schedule

The following table lists the VSU Departmental Schedule. The table by its current structure cannot be simply converted into a table of a database because it contains structural problems such as redundancy and data anomalies. As such, there is a need of normalizing the table. Normalization is a process we can use to remove design flaws from a database. A separate exercise will be allotted for this topic but for now, a design of this database is immediately presented without discussing how the design is derived.

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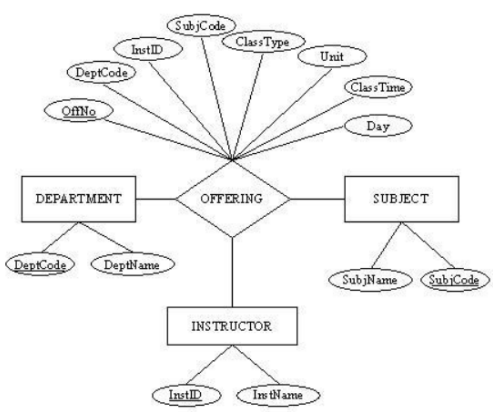
Department	**Dept. of Agricultural Engineering**						Code	DAE
Off. No.	**Subject Code**	**Descriptive Title**	**Type**	**Class Time**	**Day**	**Room**	**Units**	**Instructor**
A005	AE 30	Irrigation, drainage and meteorology	Lec	10:00-11:00	TTh	SLH	2	Fernando
A006	AE 30	Irrigation, drainage and meteorology	Lab	07:00-10:00	Th	EBWS	1	Dela Cruz
A007	AE 30	Irrigation, drainage and meteorology	Lab	07:00-10:00	T	EBWS	1	Dela Cruz
A031	AE 131n	Engineering Graphics II	Lec	10:00-11:00	F	EB201	2	Garcia
A032	AE 131n	Engineering Graphics II	Lab	01:00-04:00	F	EB107	1	Garcia
A033	AE 131n	Engineering Graphics II	Lab	01:00-04:00	Th	EB107	1	Garcia

Entity Relationships Model



Create the following Relations To Tables

- **Department**(DeptCode: char(10), DeptName: varchar(100))
- **Subject**(SubjCode: char(10), SubjName: varchar(100))
- **Instructor**(InstID: char(3), InstName: varchar(40))
- **Offering**(OffNo: char(10), SubjCode:char(10), ClassType:char(3), ClassTime:varchar(12), Day:char(7), Room:varchar(10), Units:Real, InstID:char(3), DeptCode:char(10))

Make sure that the underlined fields are set as **PRIMARY KEY** and those fields of a table that are primary key of another table are properly set as **FOREIGN KEYs**.

Example Create Statements

```
CREATE TABLE Supply
( SNum CHAR(5) NOT NULL,
SNAME VarChar(20),
STATUS TinyInt,
PRIMARY KEY (SNum) )

CREATE TABLE Part
( PNum char(6) not null,
PName varchar(20),
Weight Real,
PRIMARY KEY (PNum) )

CREATE TABLE SupplyPart
( SNum CHAR(5) NOT NULL,
PNum CHAR(6) not null,
Qty Real,
PRIMARY KEY (SNum, PNum),
FOREIGN KEY (SNum) REFERENCES Supply,
FOREIGN KEY (PNum) REFERENCES Part)
```

Data Capture

Download this data ClassSched.xls to your folder. Find ways to "automate" the insertion of these data to their appropriate tables in the database. This is to test your manipulative skills in Excel that will 'automate' the data capture.

Requirements

1. A listing of the SQL scripts in creating the tables.
2. The excel formula used in 'automating' the capture of data for each table.
3. The SQL scripts to insert the data to their corresponding tables in the database
4. A listing of data (at least 10 records) for each table.

Submit a documentation containing the scripts and its equivalent results. The submitted file should be in pdf format and with your surname as a filename.